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Current Position

Postdoctoral researcher *2025/01-2026/06*

Lab of Doct. Vanessa Stempel, Max Planck Institute for Brain Research, Frankfurt am Main, Germany.
• I analyzed Neuropixels data from the periacqueductal gray (PAG) region of freely-behaving mice, with special emphasis on population-level analysis.

Research Experience

Postdoctoral researcher *2021-2025*

Lab of Prof. Julijana Gjorgjieva, TUM School of Life Sciences, Technical University of Munich (TUM), Germany.
• Using both numerical and analytic approaches, I study how inhibitory spike-timing-dependent plasticity rules can ensure excitatory/inhibitory balance, while also generating specific connectivity structures in recurrent circuits. I also mentor students taking part in internships, with projects related to plasticity and neural dynamics, and help as a teaching assistant (Computational neuroscience course, Master of Science in Neuroengineering, TUM).

Postdoctoral researcher *2016-2021*

Lab of Dr. Ruben Coen-Cagli, Albert Einstein College of Medicine, Bronx, NY, USA
• I developed a probabilistic inference model tuned to natural image statistics, to make predictions on both mean response and trial-to-trial response variability of neurons in primary visual cortex (V1). Computationally, I reimplemented and extended the model of interest, finding new results that I used for both analytical studies and numerical simulations. In parallel to this, I worked in close collaboration with experimentalists in the laboratory of Prof. Adam Kohn, to test the theory using both existing data and new experiments. I contributed to both stimulus design and data analysis (including spike-sorting).

Ph.D. studies *2012-2016*

Group of Prof. Máté Lengyel, Computational and Biological Learning Lab (CBL), Cambridge, UK
• I built from scratch a novel optimisation procedure that operates on stabilized supralinear networks (SSNs) with multiple fixed-point attractors. The optimized networks display several relevant biological features, and can robustly store and recall memory states that correspond to population firing patterns. In my work, I modeled and tested these systems extensively, and compared them with state-of-the-art approaches used to construct attractor networks.

Research intern *2011-2012*

Lab of Prof. Fred Wolf, Max Planck Institute for Dynamics and Self-Organization, Göttingen, Germany
• Recurrent neural networks (RNNs) can produce chaotic dynamical regimes. In this project, starting from an RNN of quadratic integrate-and-fire neurons, I computed the so-called "Lyapunov vectors", which measure not only the presence of chaos, but also the direction and dimensionality of the most dynamically unstable dimensions. This allows for a more in-depth characterization of the ergodic properties of RNNs.

Publications

Structured stabilization in recurrent neural circuits through inhibitory synaptic plasticity *2024*

Dylan Festa, Claudia Cusseddu, Julijana Gjorgjieva

bioRxiv DOI: 10.1101/2024.10.12.618014

Formation and computational implications of assemblies in neural circuits *2022*

Christoph Miehl, Sebastian Onasch, **Dylan Festa**, Julijana Gjorgjieva

The Journal of Physiology,DOI: 10.1113/JP282750

Adaptation of Drosophila larva foraging in response to changes in food resources *2022*

Marina Wosniack, **Dylan Festa**, Nan Hu, Julijana Gjorgjieva, Jimena Berni

eLife,DOI: 10.7554/eLife.75826

Neuronal variability reflects probabilistic inference tuned to natural image statistics *2021*

Dylan Festa, Amir Aschner, Aida Davila, Adam Kohn, Ruben Coen-Cagli

Nature Communications,DOI: 10.1038/s41467-021-23838-x

Analog Memories in a Balanced Rate-Based Network of E-I Neurons *2014*

Dylan Festa, Guillaume Hennequin, Máté Lengyel

Advances in Neural Information Processing Systems (NeurIPS)

Teaching and Supervision Experience

Computational neuroscience course *2023,2024*

Served as teaching assistant (TA) for a university course on computational neuroscience, Technical University of Munich.

I led the exercise sessions of the course, and helped design and grade the final written exams.

Cajal summer school *2022*

Served as teaching assistant (TA) for the Cajal summer school, Champalimaud Centre for the Unknown, Lisbon, Portugal

I assisted with tutorials (first week), and then mentored two smaller groups for project work (following two weeks).

NeuroMatch Academy *2021*

Served as teaching assistant (TA) for the first Neuromatch virtual summer school in Computational Neuroscience

I assessed course materials before the course and assisted 10–15 students during the course as they worked on tutorial notebooks.

Master thesis supervision *2021*

Supervision of Master's student

I supervised the Master's student Vanshika Kapoor, for her thesis titled: "Inferring activity statistics in Stabilized Supralinear Networks"

Student supervisions *2014*

University of Cambridge, UK

During the full extent of my Ph.D. in Cambridge I tutored small groups of students in a module of the third-year course "Introduction to Neuroscience", held at the Engineering Department

Education

Ph.D. in Computational Neuroscience *2012-2016*

Group of Prof. Máté Lengyel, Computational and Biological Learning Lab (CBL), Cambridge, UK

Master's degree in Physics *2008-2011*

Università degli Studi di Pisa, Pisa, Italy

- **Final grade:** 110/110 *cum laude*. **Specialization:** Physics of Condensed Matter

Bachelor's degree in Physics *2005-2008*

Università degli Studi di Pisa, Pisa, Italy

- **Final grade:** 110/110 *cum laude*.

Diploma *2000-2005*

Liceo Scientifico D. Alighieri, Matera, Italy

- **Final grade:** 100/100.

Skills

Programming Julia | Python | MATLAB | LaTeX | OCaml | Git | Linux systems

Software Adobe Illustrator | Microsoft Office | Plexon Offline Spike Sorter

Languages Italian (native) | English (fluent) | German (beginner/intermediate, A2/B1)

Extra

- **2018-2020** Co-chair of the Einstein Postdoctoral Association (EPA), Einstein College, Bronx, NY.
- **2018** NET Impact NYC Service Corps – I volunteered as a pro bono consultant for a few months, helping a Bronx-based organization promote a K–12 school programme. The programme was based on indoor school farming using hydroponics